

# Supplementary Note 1: Methods for Information theoretic metrics for single cell RNA and T-cell receptor sequencing

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## Introduction

T cells recognize tumor antigens via their T-cell receptor (TCR) and progress through phenotypes (naive, activated, cytotoxic, exhausted). Each clonotype (cells sharing a TCR) may exhibit a distribution of phenotypes. Our model learns per-clonotype distributions and per-cell embeddings, enabling downstream measures (e.g. mutual information) over the joint distribution of the clonotypes and phenotypes to stratify therapy response. Single cell readouts from 10x with the 5' VDJ option, or even resequenced 3' libraries would be theoretically possible.

Our model, TCRI, takes as input a list of phenotype distributions for archetype clonotypes, and a gene expression vector, a clonotype assignment,

and a covariate-group index for each single cell. In this manuscript, we define archetype clonotypes as typical clonotypes that we observe in the dataset. However, they can be user defined, for instance, from a larger atlas level dataset. Covariate-groups indicate a biological sample, for example a specific timepoint or anatomical site. TCRi infers phenotype distributions at clonotype- and covariate-levels.

We use the following notation in our model:

- $N$ : number of cells in the dataset, indexed by  $i = 1, \dots, N$ .
- $D$ : number of genes measured per cell;  $P$ : number of phenotype categories.
- $L$ : dimension of gene expression latent representation per cell.
- $C$ : number of clonotypes.
- $M$ : number of clonotype-covariate-groups.
- $g : \{1, \dots, N\} \rightarrow \{1, \dots, M\}$ : mapping from a cell to its covariate-group.
- $h : \{1, \dots, M\} \rightarrow \{1, \dots, C\}$ : mapping from a covariate-group to its clonotype.
- $x_i \in \mathbb{R}^D$ : gene counts for cell  $i$ .
- $z_i \in \mathbb{R}^L$ : continuous latent embedding of cell  $i$ 's gene count computed via a variational auto encoder (VAE) architecture.<sup>1</sup>
- $z_i^\phi \in \{1, \dots, P\}$ : latent phenotype for cell  $i$ .
- $\omega_c \in \Delta^{P-1}$ ,  $c = 1, \dots, C$ : clonotype-level phenotype distributions, where  $\Delta^{P-1}$  denotes the  $P$ -simplex, that is,  $\omega_c := (\omega_{c1}, \dots, \omega_{cP})$  such that  $\omega_{cp} \geq 0$  for  $p \in \{1, \dots, P\}$  and  $\sum_{p=1}^P \omega_{cp} = 1$ .
- $\phi_m \in \Delta^{P-1}$ ,  $m = 1, \dots, M$ : covariate-level phenotype distributions.
- $\alpha > 0$ ,  $\beta > 0$ : Dirichlet concentration scales (global, local).
- $B_c$ : number of mixture components for  $p(\omega_c)$ ;  $B_z$ : number of pseudo-inputs for VampPrior on  $z_i$ .
- $\psi_b \in \Delta^{P-1}$ ,  $b = 1, \dots, B_c$ : archetype prior vectors for clonotypes.

- $u_k \in \mathbb{R}^D$ ,  $k = 1, \dots, B_z$ : pseudo-inputs for VampPrior.
- $f_{\text{cls}} : \mathbb{R}^L \rightarrow \mathbb{R}^P$ : classifier network mapping embedding to phenotype scores parameterized by  $\eta_{\text{cls}}$ .
- $\pi \in (0, 1)$ : gating weight that determines the contribution of the latent embedding and the latent covariate-level hierarchical phenotypes to the final phenotype scores.
- $\text{softmax} : \mathbb{R}^P \rightarrow \Delta^{P-1}$ ,  $\text{softmax}(\ell)_j = \exp(\ell_j) / \sum_k \exp(\ell_k)$ .
- $\eta_{\text{enc}}, \eta_{\text{dec}}$ : the encoder and decoder parameters of the VAE architecture used in gene expression modeling.<sup>2</sup>

## Generative Model

We collect the latent variables and our observables as follows:  $\mathbf{\Omega} = \{\omega_c\}_{c=1}^C$ ,  $\mathbf{\Phi} = \{\phi_m\}_{m=1}^M$ ,  $\mathbf{z} = \{z_i\}_{i=1}^N$ ,  $\mathbf{z}^\phi = \{z_i^\phi\}_{i=1}^N$ ,  $\mathbf{x} = \{x_i\}_{i=1}^N$ . The joint distribution factorizes:

$$\begin{aligned}
p(\mathbf{\Omega}, \mathbf{\Phi}, \mathbf{z}, \mathbf{z}^\phi, \mathbf{x}) &= \prod_{c=1}^C p(\omega_c) \times \prod_{m=1}^M p(\phi_m \mid \omega_{h(m)}) \times \prod_{i=1}^N \left[ p(z_i) \right. \\
&\quad \left. \times p(z_i^\phi \mid z_i, \phi_{g(i)}) \times p(x_i \mid z_i) \right].
\end{aligned}$$

### Generative model details:

#### 1. Clonotype-Level Prior (Mixture Dirichlet, $B_c$ ).

$$p(\omega_c) = \frac{1}{B_c} \sum_{b=1}^{B_c} \text{Dirichlet}(\omega_c \mid \alpha \psi_b), \quad c = 1, \dots, C.$$

This mixture prior allows each clonotype to draw from a mixture of archetypal distributions, accommodating diverse phenotype patterns. We assume  $\psi_b$  is an input to our model; this allows for user-defined priors. In practice, we use an initial cell phenotyping to compute an empirical distribution of phenotypes per clonotype. Then we group these empirical phenotype distributions and use their centroids to set each  $\psi_b$ .

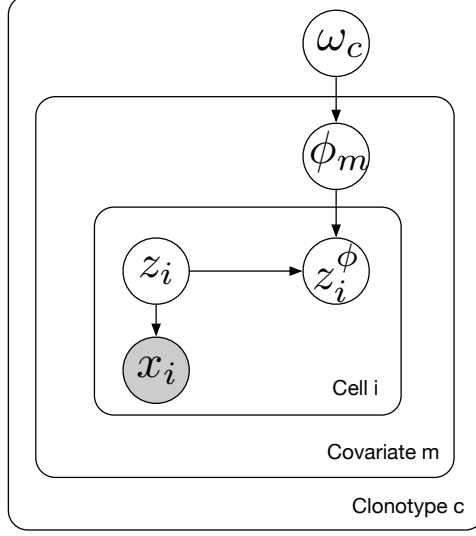


Figure 1: The probabilistic graphical model representing the model. The white and gray backgrounds denote latent and observed random variables, respectively. The generative model is summarized in listing (1).

## 2. Covariate-Level Prior (Hierarchical Dirichlet, $\beta$ ).

$$p(\phi_m \mid \omega_{h(m)}) = \text{Dirichlet}(\phi_m \mid \beta \omega_{h(m)}), \quad m = 1, \dots, M.$$

The scale  $\beta$  permits covariate-group distributions to deviate from their parent clonotype in a data-driven manner.

## 3. Latent Embedding Prior (VampPrior, $B_z$ ).

$$p(z_i) = \frac{1}{B_z} \sum_{k=1}^{B_z} q(z \mid u_k), \quad i = 1, \dots, N,$$

where  $q(\cdot)$  denotes the encoder network (c.f. the *Variational Inference* section). By approximating the aggregated posterior via learnable pseudo-inputs  $u_k$ , the VampPrior<sup>3</sup> yields a flexible, multimodal latent space that mitigates posterior collapse.<sup>4</sup>

## 4. Discrete Phenotype Likelihood. Compute

$$l_i = f_{\text{cls}}(z_i), \quad \ell_i = \pi l_i + (1 - \pi) \log \phi_{g(i)}.$$

Here  $f_{\text{cls}}$  is a neural network that outputs phenotype scores in  $\mathbb{R}^P$  and  $\pi$  is a user determined mixing weight. In our experiments, we set  $\pi = 0.5$  to give equal weight to the gene expression based and clonotype based phenotype. However, this can be adjusted depending on the down-stream task to give more weight to either source of information. Then we draw the cell-level phenotype as follows:

$$p(z_i^\phi \mid z_i, \phi_{g(i)}) = \text{Categorical}(z_i^\phi \mid \text{softmax}(\ell_i)).$$

5. **Gene Expression (ZINB).** We follow the observation model in scVI<sup>2</sup> where decoder maps each latent  $z_i$  (plus the cell’s library size) to  $(g'_i, r_i, \mu_i)$ :

$$(g'_i, r_i, \mu_i) = \text{decoder}(z_i, \log \text{library}_i; \eta_{\text{dec}}).$$

Then for each gene count  $x_{ig}$ :

$$p(x_{ig} = k) = g'_i \mathbf{1}_{k=0} + (1 - g'_i) \binom{k + r_i - 1}{k} \left( \frac{r_i}{r_i + \mu_i} \right)^{r_i} \left( \frac{\mu_i}{r_i + \mu_i} \right)^k, \quad k = 0, 1, 2, \dots$$

capturing both overdispersion ( $r_i$ ) and zero-inflation ( $g'_i$ ).

Putting these together, we have:

$$\omega_c \sim \frac{1}{B_c} \sum_{b=1}^{B_c} \text{Dirichlet}(\cdot \mid \alpha \psi_b), \quad c = 1, \dots, C, \quad (1)$$

$$\phi_m \mid \omega_{h(m)} \sim \text{Dirichlet}(\cdot \mid \beta \omega_{h(m)}), \quad m = 1, \dots, M, \quad (2)$$

$$z_i \sim \frac{1}{B_z} \sum_{k=1}^{B_z} q(\cdot \mid u_k), \quad i = 1, \dots, N, \quad (3)$$

$$z_i^\phi \mid \phi_{g(i)}, z_i \sim \text{Categorical}(\cdot \mid \text{softmax}(\ell_i)), \quad i = 1, \dots, N, \quad (4)$$

$$x_i \sim \text{ZINB}(\cdot \mid g'_i, r_i, \mu_i), \quad i = 1, \dots, N. \quad (5)$$

## Choice of Priors

- **Mixture Dirichlet ( $B_c$ ):** allows multimodal, data-informed priors for clonotype distributions, capturing heterogeneity across clones.
- **Hierarchical Dirichlet ( $\beta$ ):** balances global clonotype structure and local covariate deviations; larger  $\beta$  enforces similarity, smaller  $\beta$  permits divergence.

- **VampPrior** ( $B_z$ ): adapts the prior to the empirical latent manifold via pseudo-inputs, improving flexibility and preventing posterior collapse common with standard Normal priors.
- **Gated Categorical**: combines learned classifier signals with hierarchical prior information, enabling smooth interpolation between data-driven and prior-driven phenotype predictions.

## Variational Inference

We approximate the true posterior

$$p(\mathbf{\Omega}, \mathbf{\Phi}, \mathbf{z}, \mathbf{z}^\phi \mid \mathbf{x}; \alpha, \beta, \{\psi_b\}, \{u_k\})$$

by a mean-field variational family  $q(\cdot \mid \mathbf{x}; \Lambda)$  in the framework of variational inference.<sup>5</sup> Let

$$\Lambda = \{ (\lambda_c)_{c=1}^C, (\lambda'_m)_{m=1}^M, \eta_{\text{enc}}, \eta_{\text{cls}} \}$$

collect all variational parameters and

$$\Theta = \{ (u_k)_{k=1}^{B_z}, \eta_{\text{dec}} \}$$

collect all prior-hyperparameters and the decoder network parameters.

Using the shorthand  $q_\Lambda(\cdot \mid \mathbf{x}) := q(\cdot \mid \mathbf{x}; \Lambda)$  the variational distribution factorizes as

$$\begin{aligned} q_\Lambda(\mathbf{\Omega}, \mathbf{\Phi}, \mathbf{z}, \mathbf{z}^\phi \mid \mathbf{x}) &= \prod_{c=1}^C \text{Dirichlet}(\omega_c \mid \lambda_c) \times \prod_{m=1}^M \text{Dirichlet}(\phi_m \mid \lambda'_m) \\ &\times \prod_{i=1}^N q(z_i \mid x_i; \eta_{\text{enc}}) \times q(z_i^\phi \mid z_i, \phi_{g(i)}; \eta_{\text{cls}}). \end{aligned} \quad (6)$$

Concretely, the encoder maps the gene expression vector  $x_i$  to the parameters of a Gaussian distribution to sample  $z_i$ :

$$\begin{aligned} (\mu_i, \sigma_i) &= \text{encoder}(x_i; \eta_{\text{enc}}), \\ q(z_i \mid x_i) &= \mathcal{N}(z_i \mid \mu_i, \text{diag}(\sigma_i^2)), \end{aligned}$$

and then the phenotypes are sampled

$$\ell_i = \pi f_{\text{cls}}(z_i; \eta_{\text{cls}}) + (1 - \pi) \log \phi_{g(i)},$$

so that

$$q(z_i^\phi \mid z_i, \phi_{g(i)}) = \text{Categorical}(z_i^\phi \mid \text{softmax}(\ell_i)).$$

We jointly optimize  $\Lambda$  and  $\Theta$  by minimizing the Kullback–Leibler (KL) divergence between the variational family and the posterior distribution

$$\text{KL}(q_\Lambda(\mathbf{\Omega}, \mathbf{\Phi}, \mathbf{z}, \mathbf{z}^\phi \mid \mathbf{x}) \parallel p(\mathbf{\Omega}, \mathbf{\Phi}, \mathbf{z}, \mathbf{z}^\phi \mid \mathbf{x}; \alpha, \beta, \{\psi_b\}, \{u_k\}, \eta_{\text{dec}})).$$

This is equivalent to maximizing the evidence lower bound (ELBO):<sup>5</sup>

$$\begin{aligned} \mathcal{L}(\mathbf{x}; \Lambda, \Theta) = & \mathbb{E}_{q_\Lambda} [\log p(\mathbf{x} \mid \mathbf{z}; \eta_{\text{dec}})] \\ & + \mathbb{E}_{q_\Lambda} [\log p(\mathbf{\Omega}, \mathbf{\Phi}, \mathbf{z}, \mathbf{z}^\phi; \alpha, \beta, \{\psi_b\}, \{u_k\})] \\ & - \mathbb{E}_{q_\Lambda} [\log q_\Lambda(\mathbf{\Omega}, \mathbf{\Phi}, \mathbf{z}, \mathbf{z}^\phi \mid \mathbf{x})]. \quad (7) \end{aligned}$$

We maximize  $\mathcal{L}(\mathbf{x}; \Lambda, \Theta)$  via stochastic gradient ascent, using the reparameterization trick for continuous latents,<sup>1,6</sup> and parallel enumeration for discrete latents.<sup>7</sup>

## Inference Details

SVI (Stochastic Variational Inference) in Pyro - the probabilistic programming language used here - uses mini-batching, KL scaling for Dirichlet and discrete terms, and Adam optimization.<sup>8</sup>

In practice, we find it more computationally expedient to optimize a surrogate objective that directly penalizes misalignment between cell-level phenotype predictions and their latent covariate-level phenotype distribution. Concretely, letting  $\text{probs}_i = \text{softmax}(\ell_i)$  denote the predicted phenotype probabilities for cell  $i$ , we introduce a cell-wise KL penalty, and define:

$$\mathcal{L}_{\text{new}}(\mathbf{x}; \Lambda, \Theta) := \mathcal{L}^\#(\mathbf{x}; \Lambda, \Theta) + \gamma \sum_{i=1}^N \text{KL}(\text{probs}_i \parallel \phi_{g(i)}),$$

where  $\mathcal{L}^\#(\mathbf{x}; \Lambda, \Theta)$  is the ELBO in Equation (7), with terms that involve  $z_i^\phi$  removed.  $\gamma > 0$  controls the strength of this regularization.

## In silico Perturbation

We assess each gene’s importance by “silencing” it in the expression matrix (setting its values to zero) and measuring how much the model’s posterior phenotype distribution shifts. Concretely, for gene  $j$  we (1) form a

perturbed matrix  $X^{(j)}$  by zeroing column  $j$ ; (2) compute the unperturbed per-clonotype mean phenotype probabilities  $\bar{\phi}^{(0)}$  and, via  $S$  posterior draws on  $X^{(j)}$ , the perturbed means  $\tilde{\phi}^{(j)}$ ; and (3) define the gene’s importance  $I_j$  as the  $L_1$  distance between  $\bar{\phi}^{(0)}$  and  $\tilde{\phi}^{(j)}$ .

Here,  $\phi_{i,p}^{(0)}$  is the baseline (unperturbed) posterior probability of phenotype  $p$  for cell  $i$ , and  $\phi_{i,p}^{(s,j)}$  denotes the posterior probability of phenotype  $p$  for cell  $i$  obtained from the  $s$ -th posterior predictive draw using the perturbed matrix  $X^{(j)}$ . These draws are generated by passing  $X^{(j)}$  through the model’s encoder to sample latent  $z_i^{(s)}$  and then through the classifier with softmax normalization.

$$X_{i,g}^{(j)} = \begin{cases} 0, & g = j, \\ X_{i,g}, & g \neq j, \end{cases} \quad (8)$$

$$\bar{\phi}_p^{(0)} = \frac{1}{N} \sum_{i=1}^N \phi_{i,p}^{(0)}, \quad (9)$$

$$\bar{\phi}_p^{(s,j)} = \frac{1}{N} \sum_{i=1}^N \phi_{i,p}^{(s,j)}, \quad s = 1, \dots, S, \quad (10)$$

$$\tilde{\phi}_p^{(j)} = \frac{1}{S} \sum_{s=1}^S \bar{\phi}_p^{(s,j)}, \quad (11)$$

$$I_j = \sum_{p=1}^P |\bar{\phi}_p^{(0)} - \tilde{\phi}_p^{(j)}|. \quad (12)$$

## Benchmarks

In this section, we benchmark TCRi to measure its performance under multiple noise scenarios. We want to account for two sources of uncertainty in estimation of mutual information from single cell RNA/TCR-seq data: 1) finite sampling of cells, and 2) the phenotype assignment based on measured gene expression.

### Mutual Information Experiments

Our method, TCRi, computes the mutual information (MI) between clonotypes and phenotypes per sample. We want to design a benchmarking



scheme to compare our model against alternative methods. One idea is to use the following protocol:

1. Select a real dataset with tuples  $(X_i, C_i)$ , where  $X_i$  is the gene expression vector and  $C_i$  the clonotype label for cell  $i$ .
2. Define the target estimand

$$\text{MI}(C, \phi) = I(C; \phi)$$

where  $\phi$  denotes a discrete phenotype.

3. Retain only clonotypes with sufficient size:  $\{c : |c| > c'\}$  for some cut-off value  $c'$ .
4. Infer phenotypes via batch correction followed by clustering.
5. For each patient, randomly sample a fraction  $q\%$  of cells.
6. Run each competing method on the sub-sampled data, estimate  $\widehat{\text{MI}}$ , and report the average  $L^1$  deviation from the reference

The caveat of this approach is that we do not know the true MI. To address this, we can use semi-synthetic experiments.

## Generative Model for Semi-Synthetic Simulations

We generate data so that the true mutual information is known in advance:

$$\begin{aligned} \pi &\sim \text{Dirichlet}(\alpha_\pi), \\ \omega_c &\sim \text{Dirichlet}(\alpha_0), \quad c = 1, \dots, C, \\ z_i &\sim \text{Categorical}(\pi), \quad i = 1, \dots, N, \\ \phi_i | z_i &\sim \text{Categorical}(\omega_{z_i}), \quad i = 1, \dots, N, \\ x_i | \phi_i &\sim p(x | \phi_i), \quad i = 1, \dots, N. \end{aligned}$$

In the above,  $\pi$  denotes the relative abundance of clonotypes,  $\omega_c$  the latent phenotypic potential of each clonotype  $c$ ,  $z_i$ ,  $\phi_i$ , and  $x_i$  denote the clonotype assignment, the phenotype label, and the gene expression vector for cell  $i$  respectively.

## Using the Simulator in Model Evaluation

Given  $B$  candidate methods, we proceed:

1. Initialize simulator parameters from the target dataset.
2. Create a set of configurations, with the desired number of cells, and desired variance.
3. For each configuration, simulate  $S$  independent datasets  $D_s$  and compute the ground-truth  $\text{MI}_{D_s}$ .
4. Apply method  $b$  to each  $D_s$  to obtain  $\widehat{\text{MI}}_{b,D_s}$ .
5. Report the mean  $L^1$  error per method:

$$\text{MAE}_b = \frac{1}{S} \sum_{s=1}^S |\widehat{\text{MI}}_{b,D_s} - \text{MI}_{D_s}|.$$

## Fitting the Generative Model to a Dataset

To match the simulator to real data, we take the following steps:

1. Choose the number of phenotypes  $K$  and latent dimension  $L$ .
2. Fit nonnegative matrix factorization:  $X \approx UV^\top$ , assuming  $X_{ij} \sim \text{Poisson}([UV^\top]_{ij})$ .
3. Cluster the rows of the cell-factor matrix  $U$  using K-Means to define phenotype groups.
4. Model each cluster via a multivariate Gamma distribution, with independent components
5. Set clone frequencies  $\pi$  and phenotype potentials  $\omega_c$  by matching empirical frequencies.

## Temperature Scaling of Conditional Distributions

We apply temperature-scaling transformation on the conditional distribution  $P(\phi \mid c)$  to modulate mutual information. Given the original row-stochastic matrix

$$P(\phi \mid c), \quad c = 1, \dots, C, \quad \phi = 1, \dots, K,$$

and a temperature  $T > 0$ , we set:

$$P_T(\phi \mid c) \propto P(\phi \mid c)^{1/T},$$

then row-normalize so that  $\sum_{\phi} P_T(\phi \mid c) = 1$ . As  $T \rightarrow 0$ , each row sharpens onto its highest-mass entry (increasing MI); as  $T \rightarrow \infty$ , rows flatten toward uniform (decreasing MI). The output is the scaled conditional matrix  $P_T(\phi \mid c)$ .

### Modeling noise in the phenotype to gene expression map

One source of uncertainty in MI estimation is the uncertainty in phenotype to cell assignment based on gene expression. Some cells, with different phenotypes, may exhibit similar measured gene expression patterns. In our simulation scheme, we use Gamma distributions to represent each phenotype in the latent space. We model noise by pushing the Gamma distributions toward one another. We use a natural-parameter interpolation of Gamma distributions scheme as follows.

We introduce a tunable fuzziness parameter  $f \in [0, 1]$  and linearly interpolate the natural parameters of the  $K$  phenotype-specific multivariate Gamma distributions toward their global mean. For each phenotype  $k = 1, \dots, K$ , let

$$\alpha_k, \beta_k \in \mathbb{R}_{>0}^L$$

be its shape and rate vectors. Define the natural parameters

$$\theta_k = (\alpha_k - \mathbf{1}, -\beta_k) \in \mathbb{R}^{2L}, \quad \bar{\theta} = \frac{1}{K} \sum_{j=1}^K \theta_j.$$

We also apply a concave mapping  $g(f)$  (e.g.,  $g(f) = \sqrt{f}$ ) to amplify small  $f$ . Then interpolate:

$$\theta'_k = (1 - g(f)) \theta_k + g(f) \bar{\theta},$$

and recover updated parameters:

$$\alpha'_k = (\theta'_k)_{1:L} + \mathbf{1}, \quad \beta'_k = -(\theta'_k)_{L+1:2L}.$$

The resulting  $\{\alpha'_k, \beta'_k\}$  have controlled overlap according to  $f$ ; in the reported experiments, we use  $g(f) = \sqrt{f}$ , but any concave  $g$  on  $[0, 1]$  can be specified to produce stronger mixing at low fuzziness.

## Empirical Results

Here we study the performance of TCRi against a baseline method in inferring NMI. We use semi-synthetic experiments as defined in the above sections. To capture high confidence clonotypes, we fit the generative model to a dataset of T cells from anti-PD1 treated lung cancer patients,<sup>9</sup> subset to clonotypes with at least 60 associated T cells. This results in 9840 cells and 984 genes, and 47 clonotypes. We study the effects of number of cells  $N$ , number of phenotypes  $K$ , and the noise in gene expression to phenotype map *fuzziness*.

### Experimental Setup

To study the effects of noise in gene expression to phenotype map, we use the fuzziness parameter  $f$  (Section ). We vary the fuzziness in  $\{0.0, 0.1, \dots, 0.9\}$ . Figure 2 shows how the noise in the phenotype to gene expression map affects the separation of phenotypes. As  $f$  increases, the noise phenotypes are less discernible by their observed gene expression.

We vary  $N$  in  $\{250, 500, 1,000, 2,000, 5,000\}$ . We simulate data with  $K_{\text{true}} = 10$  phenotype categories. At inference, both TCRi and the GMM baseline are supplied with the same value of  $K \in \{8, 10, 12\}$ . For each configuration of  $(N, K, f)$ , we repeat the experiment with 10 random seeds. We compare TCRi to a Gaussian mixture model (GMM) as a baseline. We input the normalized simulated gene expression data to GMM to get clusters using various values of  $K$ . We normalize the gene expression by dividing each cell by the sum of its gene counts and multiplying it by 10,000. We then standardize each gene to have mean 0.0 and standard deviation 1.0. We use K-Means with the supplied value of  $K$  to obtain the initial phenotype assignments for TCRi. The semi-synthetic generator uses  $L = 5$  NMF factors, and TCRi is trained for a maximum of 2,000 epochs. For both TCRi and the GMM baseline, estimated NMI is computed using the observed clonotype abundances in each simulated dataset. For TCRi, we report the posterior mean NMI over 200 posterior draws. We use the absolute error between the estimated and ground-truth normalized mutual information (NMI) as the performance metric for each method.

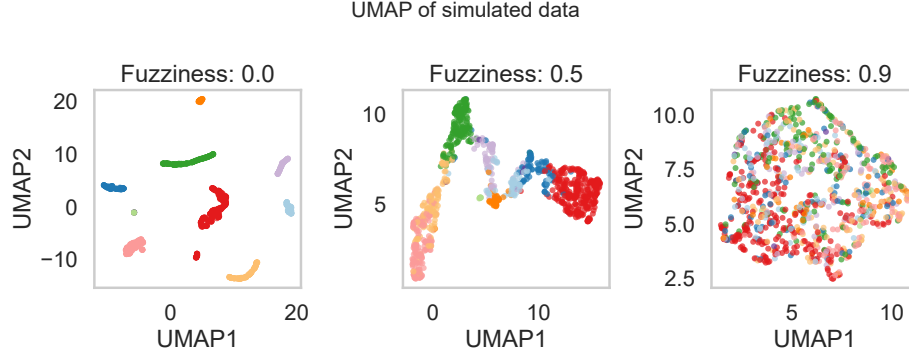


Figure 2: Modeling the noise in the phenotype to gene expression map. As the fuzziness parameter  $f$  increases, the T cell phenotypes are less separated.

### Benchmarking against a GMM

Figure 3 shows the absolute error in NMI for GMM and TCRi across configurations of  $(N, K, f)$ .

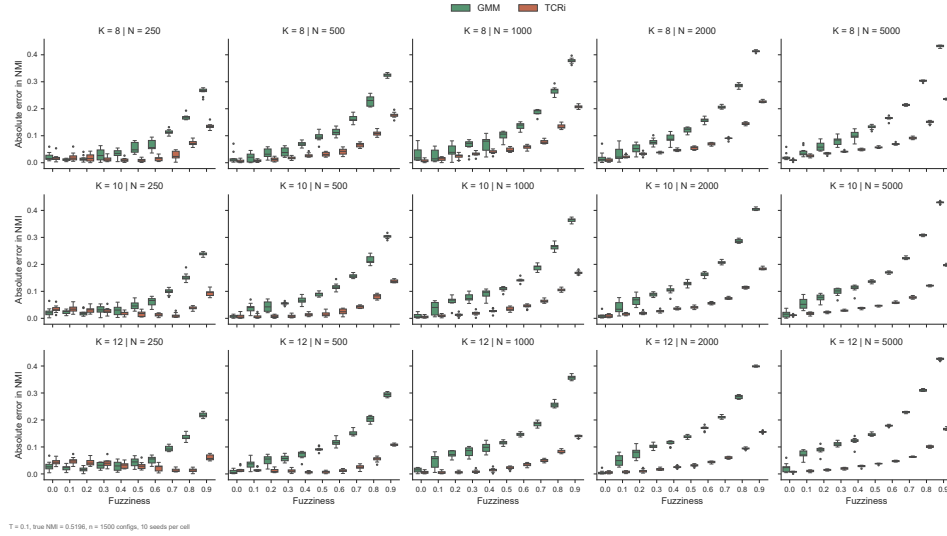


Figure 3: Absolute error between estimated and ground-truth NMI for TCRi and the Gaussian mixture model (GMM) as a function of phenotype–expression fuzziness  $f$ , the number of sampled cells  $N$ , and the supplied number of phenotype categories  $K$ . Temperature was fixed at  $T = 0.1$  (true NMI = 0.520). Boxplots summarize 10 independent simulation replicates per configuration.

As phenotype-expression fuzziness increases, absolute error increases for both methods, but TCRi generally exhibits lower error than the GMM baseline, with the largest differences observed at moderate to high fuzziness.

## Effect of $N$

We modulate the ground-truth MI using a temperature parameter. For the fitted clonotype–phenotype conditional distribution used in this experiment, temperatures of 0.1, 0.5, and 1.0 produce ground-truth NMI values of 0.520, 0.316, and 0.182, respectively, corresponding to high-, intermediate-, and low-NMI regimes.

Figure 4 shows the absolute error in TCRi-estimated NMI as the number of sampled cells, the supplied value of  $K$ , and the ground-truth NMI regime vary. Absolute error decreases with  $N$  in all three regimes. In the high-NMI regime, error approaches zero at larger sample sizes, whereas in the intermediate- and low-NMI regimes it decreases but remains appreciable

at the largest sample size. These patterns are consistent across the tested values of  $K$ .

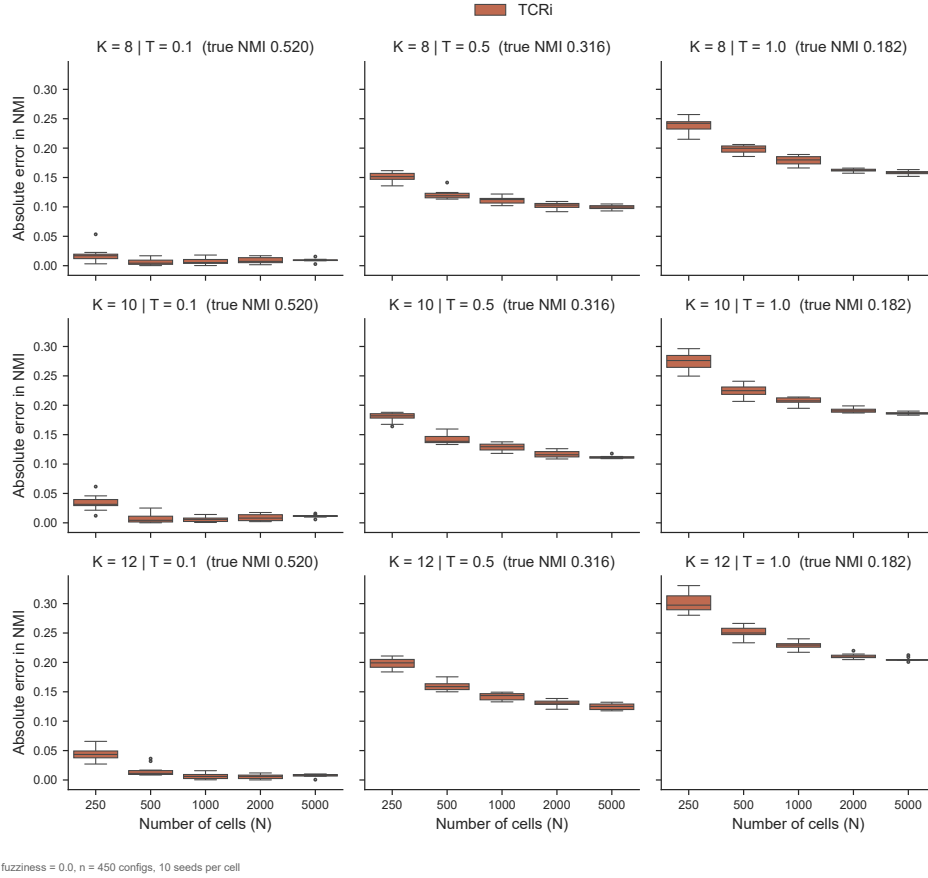


Figure 4: Absolute error between TCRi-estimated and ground-truth NMI as a function of the number of sampled cells  $N$ , the supplied number of phenotype categories  $K$ , and temperature  $T$ . Phenotype-expression fuzziness was fixed at  $f = 0$ . Temperatures of 0.1, 0.5, and 1.0 correspond to true NMI values of 0.520, 0.316, and 0.182, respectively. Boxplots summarize 10 independent simulation replicates per configuration.

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| Symbol              | Description                                        | Domain / Values         |
|---------------------|----------------------------------------------------|-------------------------|
| $N$                 | Number of cells                                    | $\mathbb{N}$            |
| $C$                 | Number of clonotypes                               | $\mathbb{N}$            |
| $M$                 | Number of clonotype-covariate-groups               | $\mathbb{N}$            |
| $P$                 | Number of phenotype categories                     | $\mathbb{N}$            |
| $D$                 | Number of genes                                    | $\mathbb{N}$            |
| $L$                 | Dimension of gene-expression latent space          | $\mathbb{N}$            |
| $x_i$               | Gene expression of cell $i$                        | $\mathbb{R}_{\geq 0}^D$ |
| $g(\cdot)$          | Covariate-group index for cell $i$                 | $\{1, \dots, M\}$       |
| $h(\cdot)$          | Clonotype for clonotype-covariate-group $m$        | $\{1, \dots, C\}$       |
| $z_i$               | Continuous latent embedding                        | $\mathbb{R}^L$          |
| $z_i^\phi$          | Discrete phenotype latent                          | $\{1, \dots, P\}$       |
| $\omega_c$          | Clonotype-level phenotype distribution             | $\Delta^{P-1}$          |
| $\phi_m$            | Covariate-level phenotype distribution             | $\Delta^{P-1}$          |
| $\alpha, \beta$     | Dirichlet concentration scales (global, local)     | $> 0$                   |
| $B_c$               | Mixture size for $\omega_c$                        | $\mathbb{N}$            |
| $B_z$               | VampPrior pseudo-inputs for $z_i$                  | $\mathbb{N}$            |
| $\psi_b$            | Archetype prior vectors for clonotypes             | $\Delta^{P-1}$          |
| $u_k$               | VampPrior pseudo-inputs for latent prior           | $\mathbb{R}^D$          |
| $l_i$               | Classifier phenotype scores for cell $i$           | $\mathbb{R}^P$          |
| $\ell_i$            | Combined phenotype scores                          | $\mathbb{R}^P$          |
| $g'_i$              | ZINB dropout probability                           | $[0, 1]$                |
| $r_i$               | ZINB dispersion (total-count) parameter            | $> 0$                   |
| $\mu_i$             | ZINB rate parameter                                | $> 0$                   |
| $\Lambda$           | All variational parameters                         | —                       |
| $\Theta$            | Prior hyperparameters & decoder network parameters | —                       |
| $\lambda_c$         | Dirichlet concentration for $q(\omega_c)$          | $\mathbb{R}_{>0}^P$     |
| $\lambda'_m$        | Dirichlet concentration for $q(\phi_m)$            | $\mathbb{R}_{>0}^P$     |
| $\mu_i$             | Encoder Gaussian mean for $z_i$                    | $\mathbb{R}^L$          |
| $\sigma_i$          | Encoder Gaussian scale for $z_i$                   | $\mathbb{R}_{>0}^L$     |
| $\psi_i$            | Variational scores for $z_i^\phi$                  | $\mathbb{R}^P$          |
| $\eta_{\text{enc}}$ | Encoder network parameters                         | —                       |
| $\eta_{\text{cls}}$ | Classifier network parameters                      | —                       |
| $\pi$               | Gating weight                                      | $(0, 1)$                |
| $\eta_{\text{dec}}$ | Decoder network parameters                         | —                       |
| $\text{library}_i$  | Library size (total counts) for cell $i$           | $> 0$                   |

Table 1: Expanded notation table with each symbol defined.